

S-DES Complete Visual Guide

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1. Inputs & Key Generation

Step	Operation Details	Visual Output
Start	Master Key (10-bit)	1010000010
1	P10 Permutation	1000001100
2	Split (Left & Right)	L: 10000 R: 01100
3	Shift Left 1 (LS-1)	L: 00001 R: 11000
4	Generate K1 (P8)	K1 → 10100100
5	Shift Left 2 (LS-2)	L: 00100 R: 00011
6	Generate K2 (P8)	K2 → 01000011

2. Encryption (Visualized)

Plaintext:

0	1	1	1	0	0	1	0
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#	Process Stage	Visual Data Flow
1	Initial Permutation (IP) <i>Rearrange input bits</i>	1 0 1 0 1 0 0 1 ⇐ Split ⇒ L_0: 1 0 1 0 R_0: 1 0 0 1
— ROUND 1 (K_1 : 10100100) —		
2	Expansion (E/P) Expand R_0 from 4 to 8 bits	Input: 1 0 0 1 ↓ Expanded: 1 1 0 0 0 0 1 1
3	XOR with K_1 (Bitwise Addition)	1 1 0 0 0 0 1 1 $\oplus$ 1 0 1 0 0 1 0 0 = 0 1 1 0 0 1 1 1
4	S-Boxes Substitution <i>Left 4 bits $\rightarrow S_0$, Right 4 bits $\rightarrow S_1$</i>	Left (0110): Row 00, Col 11 $\rightarrow$ 2 (1 0) Right (0111): Row 01, Col 11 $\rightarrow$ 3 (1 1) Combined: 1 0 1 1
5	Permutation P_4	Input: 1 0 1 1 $\xrightarrow{P_4}$ Output: 0 1 1 1
6	Update Left (L_1) $L_0 \oplus P_{4_Output}$	1 0 1 0 $\oplus$ 0 1 1 1 = 1 1 0 1
SW	SWAP Function	Old $R_0 \rightarrow$ New L: 1 0 0 1 Old $L_1 \rightarrow$ New R: 1 1 0 1
— ROUND 2 (K_2 : 01000011) —		
7	Complex Function f_K 1. Expand Right (1101) 2. XOR with K_2 3. S-Boxes	1. Exp $\rightarrow$ 11101011 2. XOR $\rightarrow$ 10101000 3. $S_0(1010) \rightarrow 2(10), S_1(1000) \rightarrow 3(11)$ Result: 1 0 1 1
8	Update Left (L_2) $L \oplus P_4(1011)$	1 0 0 1 $\oplus$ 0 1 1 1 = 1 1 1 0
9	Final Permutation (IP^{-1}) Combine L_2R_2 (1110 1101)	Ciphertext: 0 1 1 1 0 1 1 1 ✓

3. Decryption (Verification)

Ciphertext:

0	1	1	1	0	1	1	1
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#	Stage	Data Flow								
1	Initial Permutation	IP^{-1} reversed $\rightarrow$ <table><tr><td>1</td><td>1</td><td>1</td><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td></tr></table>	1	1	1	0	1	1	0	1
1	1	1	0	1	1	0	1			
— ROUND 1 (Use K_2 : 01000011) —										
2	Process	Result of $f_K(R, K_2)$ matches Encryption logic: Output is <table><tr><td>0</td><td>1</td><td>1</td><td>1</td></tr></table>	0	1	1	1				
0	1	1	1							
3	XOR Left	$L(1110) \oplus 0111 \rightarrow$ <table><tr><td>1</td><td>0</td><td>0</td><td>1</td></tr></table>	1	0	0	1				
1	0	0	1							
SW	SWAP	$L \leftarrow 1101, \quad R \leftarrow 1001$								
— ROUND 2 (Use K_1 : 10100100) —										
4	Process	Result of $f_K(R, K_1)$ matches Encryption logic: Output is <table><tr><td>0</td><td>1</td><td>1</td><td>1</td></tr></table>	0	1	1	1				
0	1	1	1							
5	XOR Left	$L(1101) \oplus 0111 \rightarrow$ <table><tr><td>1</td><td>0</td><td>1</td><td>0</td></tr></table>	1	0	1	0				
1	0	1	0							
6	Final Output	Combine $L, R \rightarrow 10101001$ Apply $IP^{-1} \rightarrow$ Plaintext: <table><tr><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td><td>1</td><td>0</td></tr></table> ✓	0	1	1	1	0	0	1	0
0	1	1	1	0	0	1	0			